

Linear Algebra Corrections

January 3, 2026

Page	Original Text	Proposed Correction	Notes	Finder
2	A vector $a \in \mathbb{R}^n$ is called bf zeros vector	A vector $a \in \mathbb{R}^n$ is called zero vector	Residual LaTeX command	Mathijs
5	Let us begin by the definition...	Let us begin with the definition...	Preposition error	Mathijs
5	there are other set of numbers	there are other sets of numbers	Pluralization	Mathijs
6	a one-dimensional array that whose elements...	a one-dimensional array whose elements...	Redundant phrasing	Mathijs
6	that the vectors a,b,c,d are unit while d is not unit.	while e is not unit.	Wrong letter	Niels
6	$\langle a, b \rangle = \langle b, a \rangle = \dots = 32, \langle a, b \rangle = 2 * 32 = 64$	$\langle 2a, b \rangle = 2 * 32 = 64$	Forgot number	Niels
7 / 10	it hold that	it holds that	Subject-verb agreement	Mathijs
11	On the other words	In other words	Idiomatic expression	Mathijs
12	play central role	play a central role	Missing article	Mathijs
Ch. 3	what a linear system of equations;	what a linear system of equations is	Incomplete sentence	Mathijs
15	lower triangular matrix. A $n \times n$ matrix A is called upper triangular ...	A $n \times n$ matrix A is called lower triangular ...	Wrong definition	Niels
16	$n \times n$ matrix A is called	$n \times n$ matrix I is called	Identity matrix notation	Joren
17	$AB = \dots = \begin{bmatrix} 32 & 27 & 17 \\ 29 & 46 & 16 \\ 90 & 72 & 47 \end{bmatrix}$	$AB = \dots = \begin{bmatrix} 32 & 27 & 17 \\ 29 & 46 & 16 \\ 90 & 78 & 47 \end{bmatrix}$	Calculation error (72 $\rightarrow$ 78)	Joren
19	the vectors $A_1, A_2, \dots$ are	the matrices $A_1, A_2, \dots$ are	Context is matrix independence	Joren
25	$b \in \mathbb{R}$	$b \in \mathbb{R}^m$	Missing dimension superscript	Joren
26 / 31	$x_1 = x_2 = 2$ and s be solution	$x_1 = -4, x_2 = 14, x_3 = -2$ and s be a solution of $As = b$	Arithmetic error in substitution	Joren
35 / 70	(see, e.g., [1].	(see, e.g., [1])	Missing closing bracket	Joren
45	$\det(A) = a_{11}C_{11} + a_{12}C_{12}$	$\det(A) = a_{11}C'_{11} + a_{12}C'_{12}$	Subscript formatting	Joren
45	Example ??	Example 57	Placeholder question marks	Joren
49	(Rule 7): ta_{12}	a_{12}	t should be removed	Joren
49	(Rule 8): a_{21}	a_{22}	Indexing error	Joren
49	(example 67): $-\det([\dots]) = 4 \det([\dots]) = -4 * 6 = -24$	$-\det([\dots]) = -4 \det([\dots]) = -4 * -6 = 24$	rekenfout	Niels
51	(Rule 8): a_{21}	a_{22}	Indexing error	Niels
51	$\begin{bmatrix} x_1 & 0 & 0 \\ x_2 & a_1 & 0 \\ x_3 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} x_1 & 0 & 0 \\ x_2 & 1 & 0 \\ x_3 & 0 & 1 \end{bmatrix}$	Variable should be 1	Joren
52	c_{33}	C_{33}	Cofactor notation	Joren
53	$Ax = \dots = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$	$Ax = \dots = \lambda \begin{bmatrix} -4 \\ 1 \end{bmatrix}$	Missing scalar λ	Joren
54	Aw	Ax	Consistent variable usage	Joren

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54	B (with diagonal 5, 5)	B (with diagonal 4, 4)	Consistency with text below	Joren
55	$\lambda_2 =$ and $\lambda_3 = 3$	$\lambda_2 = 2$ and $\lambda_3 = 3$	Missing value	Joren
56	prodict / b	product / B	Spelling/Case	Joren
57	$7 \geq \frac{\langle x, Ax \rangle}{\ x\ ^2} \leq 2$	$7 \geq \frac{\langle x, Ax \rangle}{\ x\ ^2} \geq 2$	Inequality direction	Joren
58	einvalues	eigenvalues	Spelling	Mathijs
64 / 65	Nondiagonalizable matices	Nondiagonalizable matrices	Spelling	Mathijs
65	Checking diabolizablility	Checking diagonalizability	Spelling	Mathijs
65	eignvectors / eigenvlaues	eigenvectors / eigenvalues	Spelling	Mathijs
68	$\lambda_1 = 1$ and $\lambda_1 = 0$	$\lambda_1 = 1$ and $\lambda_2 = 0$	Wrong index	Niels
69	$b\mathbb{R}^n$	$b \in \mathbb{R}^n$	Missing symbol	Joren
71	perpendicular to to V	perpendicular to V	Word repetition	Joren
72	prependicular	perpendicular	Spelling	Mathijs
73	[...] <i>projection of a onto the line through b, [...]</i>	[...] projection of b onto the line through a, [...]	Switched vectors a and b	Torben
73	$\dim(C(A^T)) = \dim(C(A))$	$\dim(C(A^T)) = \dim(C(A))$	Missing brackets	Joren
75	some times	sometimes	Compound word	Mathijs
75	$\langle b_1, a_1 \rangle - \langle b_1, a_1 \rangle$	$\langle b_1, a_2 \rangle - \langle b_1, a_2 \rangle$	Indexing error	Joren
78	prependicular	perpendicular	Spelling (occurred multiple times)	Mathijs
79	eigenvalues are positive	eigenvalues are strictly positive	Definitional precision	Joren
80	a one solution / guaranttes	one solution / guarantees	Grammar/Spelling	Joren
82	$A \in \mathbb{S}^{n \times n}$	$A \in \mathbb{R}^{n \times n}$	Generalize from symmetric	Joren
82	$a_{ii} \geq \sum$	$a_{ii} > \sum$	Strict dominance	Joren
82	if 0 is its smallest eigenvalue	if all eigenvalues are ≥ 0	Definition of PSD	Joren
83	orthonomally	orthonormally	Spelling	Joren
85	$A^T A$ contains 8	$A^T A$ contains 0	Matrix calculation error	Joren
86	$\sigma_i > 0$	$\sigma_i \geq 0$	SVs can be zero	Joren
86	$\sigma_i u_i$	$\sigma_i^2 u_i$	Missing square in definition	Joren
88	$n - 1$ zero columns	$n - r$ zero columns	Rank notation error	Joren
88	- the eigenvalues v_j ...	- the eigenvectors v_j ...	fout begrip	Niels
89	$\sigma_1 = 3, \sigma_1 = 2, \sigma_1 = 1$	$\sigma_1 = 3, \sigma_2 = 2, \sigma_3 = 1$	Subscript sequence	Joren
91	grapg	graph	Spelling	Mathijs
92	indeces	incidence	Correct term: Incidence Matrix	Joren
93	underdetermind	underdetermined	Spelling	Mathijs
93	$\partial x_1 1$	∂x_1	Typo in partial derivative	Joren
94	eigenvalues	eigenvectors	Context refers to vectors	Joren